2024827

I BE Examination May 2024
(For Computer Science (B) Students)
2MERC3-ENGINEERING DRAWING

Time 3 Hours Max. Marks: 60

Note: All the questions are compulsory. Assume suitable data if necessary. Internal choice are given in each question.

Attempt Any Two from question no.1.

Q.1	A	A 3.2 cm long line represents a length of 4 metres. Extend this line to measure lengths upto 25 metres and show on it units of metre and 5 metres. Show the length of 1 7 metres on this line.	6
	B	Draw a Vernier scale of RF = 1 / 25 to read centimetres up to 4 meters and on it, show lengths 2.39 m and 0.91 m.	6
	C	A ball thrown in air attains 100 m height and covers horizontal distance 150 m on ground. Draw the path of the ball.	6
Q.2	A	Line AB 75 mm long makes 45° inclination with VP while its FV makes 55°. End A is 10 mm above HP and 15 mm in front of VP. If line is in 1st quadrant draw its projections and find it's inclination with HP.	6
	B	The front view of a 125 mm long line PQ measures 75 mm and its top view measures 100 mm. Its end Q and the mid-point M are in the first quadrant, M being 20 mm from both the planes. Draw the projections of the line PQ.	6
		OR	
	C	A person observes two objects, A & B, on the ground, from a tower, 15 m high, at the angles of depression 30° & 45°. Object A is due North-West direction of observer and object B is due West direction. Draw projections of situation and find distance of objects from observer and from tower also.	12
Q.3	A	Draw the projections of a pentagonal prism of 25 mm base side and 50 mm long axis, resting on one of its rectangular faces on the HP with the axis inclined at 45 degree to the VP.	6
	B	Draw the development of lateral surfaces of a pentagonal pyramid pentagonal pyramid side of base 25 mm and height 70 mm, when a through hole of diameter 30 mm is cut parallel to base and its centre is 17 mm above it. Pentagon is resting on HP and one of its side of base is parallel to V P.	6
		OR	
	C	A pentagonal prism, 30 mm base side & 50 mm axis is standing on HP on its base whose one side is perpendicular to VP. It is cut by a section plane 45° inclined to HP, through midpoint of axis. Draw FV, sectional TV & sectional Side view. Also draw true shape of section and Development of surface of remaining solid.	12

Attempt Any Two from Question no. 4.

Q.4 A What is the isometric projection? What is the difference between isometric view and projection? 6

B The pictorial view of the solid is given in figure 1. Draw its front view, top view and side view. 6

C Draw isometric view of a sphere of diameter 15 mm resting on a cube of side 20 mm. side of the cube are equally inclined to vertical plane. 6

Attempt Any Two from Question no. 5.

Q.5 A A semi-circular plate of 80 mm diameter has its straight edge on the VP and inclined at 30 degrees to the HP, while the surface of the plate is inclined at 45 degrees to the VP Draw the projection of the plate. 6

B Write down the difference between First and Third Angle Projection. 6

C A square prism of base 50 mm side and height 125 mm stands on the ground with a side of the base inclined at 30° to the V.P. It is penetrated by a cylinder, 50 mm diameter and 125 mm long, whose axis is parallel to both the H.P. and the V.P. and bisects the axis of the prism. Draw the projections showing fully the curves of intersection. 6

Figure-1